

- d) Two parallel 0.25 m lengths of the elastic cord are used in the catapult as shown in Fig. 3(c).

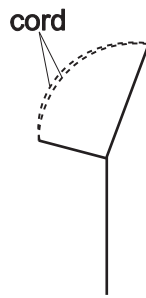


Fig. 3(c)

What is the maximum speed at which the mass can now be propelled?

[2 marks]

- A $\frac{1}{2} V_{\max}$
- B $\frac{1}{\sqrt{2}} V_{\max}$
- C $V_{\max}$
- D $\sqrt{2} V_{\max}$
- E $2V_{\max}$

SHOW YOUR REASONING IN THE SPACE PROVIDED BELOW.

Answer:

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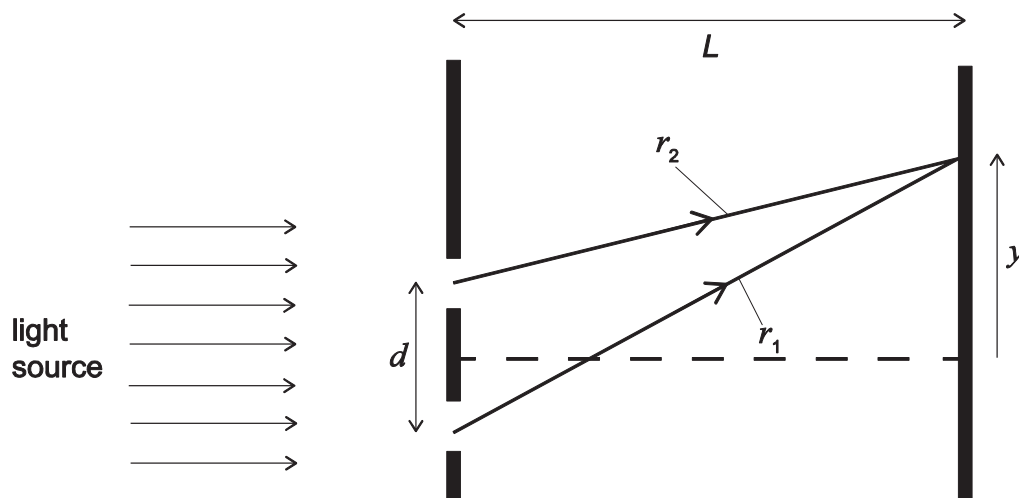
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- 4 The diagram shows the geometry for two slit diffraction of light, with the slits on the left and the viewing screen on the right; $d = 800 \text{ nm}$, $L = 1 \text{ m}$ and the speed of light is $3 \times 10^8 \text{ m s}^{-1}$.



- a) The pair of slits is illuminated by laser light of wavelength $\lambda = 600 \text{ nm}$.

Which of the following statements are correct (where n is an integer)?

[2 marks]

- 1 Points of maximum brightness on the screen occur where the distances r_1 and r_2 differ by $n\lambda$.
- 2 Points of maximum brightness on the screen occur where the distances r_1 and r_2 differ by $(n + \frac{1}{2})\lambda$.
- 3 Points of minimum brightness on the screen occur where the distances r_1 and r_2 differ by $(n + \frac{1}{2})\lambda$.
- 4 For a diffraction pattern to appear, the light from the two slits must be coherent.
- 5 The maxima are all of equal brightness.

- A 1 and 4 only
 B 1, 3 and 4 only
 C 1, 3 and 5 only
 D 1, 4 and 5 only
 E 2 and 4 only

SHOW YOUR REASONING IN THE SPACE PROVIDED ON THE NEXT PAGE.

[illegible]

- b)** A thin piece of transparent material, thickness 300 nm and in which the speed of light is half that in air, is now placed immediately behind one of the two slits.

Which one of the following statements is correct?

[3 marks]

- A** The diffraction pattern is unchanged.
- B** The diffraction pattern disappears because the light from the two slits is no longer coherent.
- C** The diffraction pattern disappears because the light from the two slits is no longer in phase.
- D** The complete diffraction pattern shifts in the y direction.
- E** Each maximum is replaced by two because the material alters the wavelength of the light coming from it.

SHOW YOUR REASONING IN THE SPACE PROVIDED BELOW.

[illegible]

- c) A radio transmitter transmits a signal at 600 MHz to a receiver 1 km away. In an attempt to double the strength of the signal at the receiver, a second antenna is added at the transmitter, 1 m away alongside the original one, and fed by the same signal. It is suggested that, instead of improving reception, diffraction effects might actually make reception much worse.

Which of the following statements is correct?

[3 marks]

- A** Diffraction effects would not be a problem because light and radio are different types of wave.
- B** Diffraction effects would not be a problem because the waves are too low frequency to produce diffraction effects.
- C** Diffraction effects would not be a problem as the transmitting antennas are too far apart to produce diffraction effects.
- D** Diffraction effects will occur, but the maxima would be sufficiently close together that this would not be a problem.
- E** Diffraction effects could be a problem because the distance between the transmitting antennas is comparable to the wavelength.

SHOW YOUR REASONING IN THE SPACE PROVIDED BELOW.

Answer:

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